

Skills Summary

From Riemann Sums to the Fundamental Theorem

Use this reference while completing your worksheet or when studying.

Skill 1 — Riemann Sums as Estimates

Recipe. $\Delta x = \frac{b-a}{n}$, then add $f(x_k)\Delta x$ over the chosen sample points: left endpoints, right endpoints, or midpoints.

Which way it leans: for an increasing f a left sum underestimates and a right sum overestimates; for a decreasing f it is the other way round.

Example: Right Riemann sum with 4 subintervals for $\int_0^4 x^2 dx$.

Step 1. The interval is asked for in four equal pieces and the sample points are the right ends, so build the widths first and then add the heights.

Step 2. $\Delta x = 1$ and the right endpoints are 1, 2, 3, 4, so the sum is $1 + 4 + 9 + 16$.

$\Rightarrow$ 30

Try it: Right Riemann sum with 3 subintervals for $\int_0^3 x^2 dx$.

check: 14

Skill 2 — Accumulated Totals from a Rate

Rule. If $r(t)$ is a rate, then $\int_a^b r(t) dt$ is the *total amount* accumulated between a and b — evaluate an antiderivative at the two limits and subtract.

Units multiply: a rate in litres per minute times minutes leaves litres. A negative rate subtracts from the total.

Example: Find $\int_0^5 (2t + 1) dt$.

Step 1. The question asks how much accumulated in total, not how fast, so an antiderivative is needed and both limits get substituted.

Step 2. $F(t) = t^2 + t$, so the value is $(25 + 5) - 0$.

$\Rightarrow$ 30

Try it: Find $\int_0^4 (2t + 3) dt$.

check: 28

Skill 3 — What the FTC Lets You Conclude

Rule. For $F(x) = \int_a^{u(x)} f(t) dt$,

$$F'(x) = f(u(x)) \cdot u'(x).$$

So the *sign of the integrand* decides where F rises and falls, and a sign change of f marks a local extremum of F — all without ever computing F itself.

Example: $G(x) = \int_1^{x^2} (t + 2) dt$. Find $G'(3)$.

Step 1. The upper limit is a function of x , so the theorem supplies the integrand at that limit and the chain rule supplies its derivative.

Step 2. $G'(x) = (x^2 + 2) \cdot 2x$, so $G'(3) = 11 \cdot 6$.

$$\Rightarrow \mathbf{G'(3) = 66}$$

Try it: $K(x) = \int_1^{x^2} 3t dt$. Find $K'(2)$.

check: $K'(2) = 48$